

Enhancing Residential Property Valuation through Spatiotemporal Multi-Task Quantile Regression Framework

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Abstract—Accurate and forward-looking residential price indices are a critical input for homebuilders, institutional investors, and lenders. This paper presents a spatio-temporal forecasting framework that operates at the ZIP×month level and produces probabilistic forecasts for one- and two-month-ahead housing price indices.

Starting from raw transaction data scraped from public listing portals, where each dwelling is associated with a historical listing/sale event sequence, we construct a hierarchically pooled ZIP×month price index that combines median sale and listing prices with county and state backfilling. On top of this index we build a multi-task tabular neural network that jointly predicts log returns for two horizons using quantile regression, geo-tiling features, state-aware winsorization, and reliability weighting based on transaction depth and history length.

On an internal dataset of US residential properties, the proposed model achieves holdout Weighted Absolute Percentage Error (WAPE) of 4–5% with $R^2 > 0.995$ across multiple rolling backtests. When evaluated on a third, out-of-sample *company index* covering 486 ZIP×month cells across seven states, the model attains state-level MAE between \$80 and \$150, and ZIP-level MAE frequently below \$100, demonstrating that the learned spatio-temporal representations transfer effectively to unseen geographies.

Index Terms—House price forecasting, ZIP-level index, quantile regression, tabular neural networks, spatio-temporal modeling.

I. INTRODUCTION

Residential real estate is one of the largest and slowest-moving asset classes, yet local market conditions can change abruptly. Homebuilders and institutional investors must continuously update beliefs about:

- the **level** of prices in each local market (e.g., ZIP code);
- the **short-horizon trajectory** of those prices over the next 1–2 months;
- the **uncertainty** and tail risk around those trajectories.

Traditional models often operate either at a coarse spatial resolution (e.g., metro-level repeat-sales indices) or at the individual property level (hedonic regressions). In contrast, this work focuses on the ZIP×month panel as the primary forecasting unit. This resolution is **fine enough** to reflect neighborhood trends and inventory shifts, yet **aggregated enough** to stabilize idiosyncratic property noise.

A. Problem Setting

We assume access to a large, web-scraped dataset from an online real-estate platform, where each record corresponds to a dwelling and includes:

- structured attributes (bedrooms, bathrooms, square footage, lot size, year built, home type);
- geographic identifiers (ZIP code, county, state, latitude, longitude);
- a **historical listing/sale event sequence** for that dwelling, i.e., a time-ordered list of listing, price change, and sale events with timestamps and prices;
- limited macro and market indicators (mortgage rate, unemployment, hotness scores).

Each dwelling is assigned a **unique property identifier** that allows us to join static attributes with its event sequence. From this raw data we derive:

- 1) a **ZIP×month price index** $P_{z,t}$, approximating typical transaction price in ZIP z during month t ;
- 2) **future labels** $R_{z,t}^{(1)}$ and $R_{z,t}^{(2)}$, defined as **log returns** of the index at horizons of one and two months;
- 3) a rich set of **temporal, macroeconomic, and spatial features** per ZIP×month cell.

Our goal is to learn a mapping

$$f_\theta : (\text{features at } (z, t)) \mapsto \{\hat{Q}_\tau^{(1)}(z, t), \hat{Q}_\tau^{(2)}(z, t)\}_{\tau \in \{0.1, 0.5, 0.9\}},$$

where $\hat{Q}_\tau^{(1)}$ and $\hat{Q}_\tau^{(2)}$ are **predicted quantiles** of the one- and two-month ahead log-return distributions. These quantiles induce both **point forecasts** (via the median) and **prediction intervals** (via the 10th and 90th percentiles).

B. Contributions

The main technical contributions of this paper are:

- **Hierarchically pooled ZIP index:** we construct a robust ZIP×month index that combines median sale and listing prices within a ZIP and backfills missing cells via county-level and state-level medians.
- **Reliability-weighted multi-task quantile network:** we propose a tabular neural architecture that jointly predicts

one- and two-month ahead log returns using quantile regression, with state-aware winsorization and data-driven reliability weights.

- **Spatio-temporal generalization study:** beyond a main evaluation on the scraped dataset, we test the learned model on a third **company index** covering 486 ZIP×month cells across seven US states, and analyze performance by state, county, and ZIP.

II. RELATED WORK

House price prediction has been extensively studied using both classical and modern machine learning approaches. Recent surveys [1] compare regression trees, support vector machines, neural networks, and ensemble methods for predicting individual property prices. Zillow’s *Neural Zestimate* [2] develops a highly engineered deep learning stack that combines image and tabular models to estimate single-home values at national scale.

Spatio-temporal modeling of real estate indices often relies on autoregressive and panel data models, sometimes augmented with neighborhood graphs or spatial lag terms [3]. Several works have explored graph and deep learning formulations for price forecasting at the level of grid cells or neighborhoods, emphasizing spatial dependencies and temporal dynamics.

Tabular deep learning has recently achieved strong performance on structured data using architectures that blend embeddings for categorical features with MLP trunks and attention modules [4], [5], [6]. Our work follows this line by adopting a **tabular multi-task quantile network** tailored to ZIP×month indices.

Finally, there is a growing ecosystem of **data-driven real estate analytics platforms** such as Reventure [7] that publish market-level dashboards and composite indicators. However, details of their modeling pipelines are rarely public. Our contribution is a **fully transparent, end-to-end description** of a ZIP×month forecasting workflow starting from raw transaction event sequences.

III. DATA ENGINEERING AND INDEX CONSTRUCTION

A. Event-Level Flattening of Transaction Histories

Each dwelling is associated with a **historical listing/sale event sequence**: a list of event dictionaries describing listings, price changes, status updates, and final sales, each with timestamps and prices. We construct an **event-level panel** via the following steps:

- 1) **Sequence parsing and normalization:** for each property, we parse its event sequence into a structured list and normalize date, time, price, and event-type fields.
- 2) **Event explosion:** we explode each sequence into one row per event, preserving the index of the event within the sequence.
- 3) **Field extraction:** for each event we extract:
 - calendar date and Unix timestamp;
 - event type (e.g., “listed for sale”, “price change”, “sold”);

- quoted price and, when available, price per square foot;
- a rental flag and multiple listing service (MLS) metadata.

- 4) **Canonical timestamp:** we define a canonical event time by preferring the high-resolution Unix timestamp and falling back to the calendar date when needed.
- 5) **Within-property ordering:** events for each unique property identifier are sorted by date, canonical time, and sequence index, and we compute the time since previous and first events.

This procedure yields a **clean, chronologically ordered event panel** of roughly half a million listing and sale events, each linked back to its unique property identifier.

B. ZIP×Month Aggregation

For each event we define a canonical **event date** and restrict attention to non-rental events from January 2022 onward. Each event is assigned to a **year-month bucket**

$$t = \text{floor_to_month}(\text{event date}).$$

Events are mapped into two broad categories:

- **Listing-like events** (initial listings, relistings, price changes) representing asking prices.
- **Sold-like events** (sales, closings) representing realized transaction prices.

For each key (z, c, s, t) , where z denotes ZIP, c county, s state, and t year-month, we compute:

$$\begin{aligned} n_{z,t}^{\text{sold}} &= \#\{\text{sold-like events with price}\}, \\ n_{z,t}^{\text{list}} &= \#\{\text{listing-like events with price}\}, \\ P_{z,t}^{\text{sold}} &= \text{median}(\text{sold-like prices}), \\ P_{z,t}^{\text{list}} &= \text{median}(\text{listing-like prices}). \end{aligned}$$

We then define a **raw ZIP-level index**

$$P_{z,t}^{\text{raw}} = \begin{cases} P_{z,t}^{\text{sold}}, & \text{if } n_{z,t}^{\text{sold}} > 0, \\ P_{z,t}^{\text{list}}, & \text{otherwise.} \end{cases} \quad (1)$$

C. Hierarchical Pooling and Relative Indices

Because many ZIP×month cells have few or zero transactions, we **borrow strength from higher geographies**. For each county c and state s , month t , we compute:

$$\begin{aligned} P_{c,t}^{\text{cty}} &= \text{median}_{z \in c}(P_{z,t}^{\text{raw}}), \\ P_{s,t}^{\text{st}} &= \text{median}_{c \in s}(P_{c,t}^{\text{cty}}). \end{aligned}$$

The **hierarchically pooled effective index** is then

$$P_{z,t} = \text{coalesce}(P_{z,t}^{\text{raw}}, P_{c(z),t}^{\text{cty}}, P_{s(z),t}^{\text{st}}), \quad (2)$$

where *coalesce* returns the first non-missing argument.

We also construct **relative indices**:

$$\text{RelCty}_{z,t} = \frac{P_{z,t}}{P_{c(z),t}^{\text{cty}}}, \quad (3)$$

$$\text{RelSt}_{z,t} = \frac{P_{z,t}}{P_{s(z),t}^{\text{st}}}, \quad (4)$$

which encode whether ZIP z is **cheap or expensive relative to its county and state** in month t and later serve as cross-sectional features.

D. Reliability Weights

Not all ZIP $\times$ month observations are equally reliable. We define a **reliability weight** $w_{z,t}$ that depends on:

- the **history length** of the ZIP up to time t ;
- the **transaction depth** in that month.

Let $n_{z,t}^{\text{tx}} = \max(n_{z,t}^{\text{sold}}, n_{z,t}^{\text{list}})$ and let $A_{z,t}$ denote the number of months since the ZIP first appeared in the index. We set

$$w_{z,t}^{\text{hist}} = \frac{\log(1 + \min(A_{z,t}, 24))}{\log(1 + 12)}, \quad (5)$$

$$w_{z,t}^{\text{tx}} = \max\left(\frac{\log(1 + n_{z,t}^{\text{tx}})}{3}, 0.2\right), \quad (6)$$

$$w_{z,t} = \min(w_{z,t}^{\text{hist}} \cdot w_{z,t}^{\text{tx}}, 1.0). \quad (7)$$

These weights later modulate the loss for the one-month horizon, allowing the network to **focus learning on well-observed ZIPs** while still exploiting information from sparser areas.

IV. FEATURE ENGINEERING AND TARGETS

A. Temporal Features and Seasonality

For each ZIP $\times$ month we define a **global month index**

$$\text{GMI} = 12 \cdot (\text{year} - 2015) + (\text{month} - 1), \quad (8)$$

and construct smooth **seasonal embeddings**:

$$\sin_{12} = \sin\left(\frac{2\pi}{12} \text{GMI}\right), \quad \cos_{12} = \cos\left(\frac{2\pi}{12} \text{GMI}\right), \quad (9)$$

$$\sin_6 = \sin\left(\frac{2\pi}{6} \text{GMI}\right), \quad \cos_6 = \cos\left(\frac{2\pi}{6} \text{GMI}\right). \quad (10)$$

These trigonometric features allow the network to **represent annual and semi-annual cycles** without hand-crafted month dummies.

B. Index Lags, Momentum, and Volatility

We construct a standard set of **time-series covariates** from the index level:

$$P_{z,t-1}, \quad P_{z,t-3}, \quad P_{z,t-6}, \quad (11)$$

$$\Delta_{z,t}^{(1)} = \frac{P_{z,t}}{P_{z,t-1}} - 1, \quad \Delta_{z,t}^{(3)} = \frac{P_{z,t}}{P_{z,t-3}} - 1, \quad \Delta_{z,t}^{(6)} = \frac{P_{z,t}}{P_{z,t-6}} - 1, \quad (12)$$

and **rolling volatility** features:

$$\sigma_{z,t}^{(3)} = \text{std}(P_{z,t-2:t}), \quad (13)$$

$$\sigma_{z,t}^{(6)} = \text{std}(P_{z,t-5:t}). \quad (14)$$

These covariates summarize **momentum and risk** at multiple horizons.

C. Geo-Tiling and Categorical Features

True H3 indexing is not available in our constrained environment, so we use an **approximate H3-style geo-tiling**. We aggregate property coordinates to ZIP $\times$ month medians and quantize latitude and longitude to three decimal places, defining

nested geo-cells (coarse and fine spatial bins). Together with ZIP, county, and state identifiers, these categorical features provide the network with a **multi-scale spatial discretization** analogous to positional encodings.

D. Macro Lags

For each available macro variable X_t (e.g., monthly mortgage rate, unemployment rate), we generate level and difference features:

$$X_t^{\text{L1}} = X_{t-1}, \quad X_t^{\text{D1}} = X_t - X_{t-1}, \quad (15)$$

$$X_t^{\text{L3}} = X_{t-3}, \quad X_t^{\text{D3}} = X_t - X_{t-3}, \quad (16)$$

$$X_t^{\text{L12}} = X_{t-12}, \quad X_t^{\text{D12}} = X_t - X_{t-12}. \quad (17)$$

These covariates allow the model to **capture sensitivity to shifts in financing costs and local labor markets**.

E. Forecasting Targets: Log Returns

Rather than forecasting price levels directly, we model **log returns of the ZIP index**. For each (z, t) we define future levels

$$P_{z,t}^{(1)} = P_{z,t+1}, \quad (18)$$

$$P_{z,t}^{(2)} = P_{z,t+2}, \quad (19)$$

and then compute

$$R_{z,t}^{(1)} = \log(1 + P_{z,t}^{(1)}) - \log(1 + P_{z,t}), \quad (20)$$

$$R_{z,t}^{(2)} = \log(1 + P_{z,t}^{(2)}) - \log(1 + P_{z,t}). \quad (21)$$

Working in log space stabilizes the scale across cheap and expensive ZIPs and converts multiplicative changes into **approximately additive** quantities, which empirically yields **better-behaved label distributions** and more stable optimization.

F. State-Aware Winsorization

To mitigate extreme outliers, we apply **state- and month-specific winsorization**. Within each (state, month) group we compute the interquartile range (IQR) of $R^{(1)}$ and $R^{(2)}$ on the training set and define lower/upper fences:

$$\ell^{(1)} = Q_1 - 1.5 \cdot \text{IQR}, \quad h^{(1)} = Q_3 + 1.5 \cdot \text{IQR}, \quad (22)$$

$$\ell^{(2)} = Q_1 - 3.0 \cdot \text{IQR}, \quad h^{(2)} = Q_3 + 3.0 \cdot \text{IQR}. \quad (23)$$

Labels are then clipped:

$$\tilde{R}_{z,t}^{(1)} = \text{clip}(R_{z,t}^{(1)}, \ell^{(1)}, h^{(1)}), \quad (24)$$

$$\tilde{R}_{z,t}^{(2)} = \text{clip}(R_{z,t}^{(2)}, \ell^{(2)}, h^{(2)}). \quad (25)$$

Aggressive winsorization for the one-month horizon reflects its shorter horizon and higher sensitivity to noise, while **more lenient clipping for the two-month horizon** preserves medium-term swings.

V. MULTI-TASK TABULAR QUANTILE NETWORK

A. Architecture

The forecasting model is a **tabular neural network** with shared trunk and two task-specific heads:

- **Numeric features** (index lags, momentum, volatility, seasonal embeddings, macro lags, reliability weight, relative indices) are standardized using a StandardScaler fit on the training set.
- **Categorical features** (ZIP, county, state, geo-cells) are mapped to integer IDs and fed into learned **embedding**

tables; unseen categories map to a shared “unknown” index.

Embeddings for all categorical features are concatenated and then concatenated with numeric features to form an input vector $x \in \mathbb{R}^D$. The shared trunk is a **deep MLP**:

$$h_0 = x, \quad (26)$$

$$h_{l+1} = \text{Dropout}(\text{ReLU}(W_l h_l + b_l)), \quad l = 0, \dots, L-1, \quad (27)$$

with hidden sizes of 384 units and dropout rate 0.15. Two linear heads map h_L to quantile predictions for horizons 1 and 2:

$$\hat{q}^{(h)} = W^{(h)} h_L + b^{(h)}, \quad h \in \{1, 2\},$$

where $\hat{q}^{(h)} \in \mathbb{R}^3$ corresponds to quantiles $\tau \in \{0.1, 0.5, 0.9\}$.

B. Loss Function

For each sample, horizon h , and quantile τ , the **pinball loss** is defined as

$$\rho_\tau(u) = \max(\tau u, (\tau - 1)u), \quad u = y - \hat{y}_\tau.$$

The quantile loss for horizon h is

$$L_{\text{pin}}^{(h)} = \frac{1}{3} \sum_{\tau \in \{0.1, 0.5, 0.9\}} \mathbb{E}[\rho_\tau(y^{(h)} - \hat{y}_\tau^{(h)})],$$

and we add an **L1 median stabilization term**

$$L_{\text{med}}^{(h)} = \mathbb{E}[|y^{(h)} - \hat{y}_{0.5}^{(h)}|].$$

The total loss for horizon h is

$$L^{(h)} = L_{\text{pin}}^{(h)} + \lambda_{\text{med}} L_{\text{med}}^{(h)},$$

with $\lambda_{\text{med}} = 0.5$.

For the one-month horizon we further **reweight the loss by reliability**:

$$\tilde{L}^{(1)} = \mathbb{E}[w_{z,t} \cdot L_{z,t}^{(1)}],$$

where $w_{z,t}$ is defined in (7). The final objective is

$$L_{\text{total}} = \tilde{L}^{(1)} + L^{(2)}.$$

C. Training Schedule

We train using the **AdamW optimizer** with learning rate 2×10^{-3} , weight decay 10^{-4} , and batch size 2048. Early stopping is driven by the sum of validation MAE for the two horizons, with patience of eight epochs. In our main experiment, early stopping selects a model around epoch 15.

D. Algorithmic Summary

The full training loop is summarized in **Algorithm 1**, using the same style as the reference CTGAN+LLM algorithm.

VI. EVALUATION PROTOCOL

A. Data Splits

For the main dataset, we keep ZIP×month rows with valid labels and non-missing index levels. Let t_{max} be the latest month in the panel. We define:

- **Holdout period:** last two months ($t_{\text{max}}, t_{\text{max}} - 1$);
- **Validation period:** preceding two months ($t_{\text{max}} - 2, t_{\text{max}} - 3$);
- **Training period:** all earlier months.

Algorithm 1 ZIP×Month TAB-MTL Training Loop

Require: Raw property records with historical listing/sale event sequences; hyperparameters for index construction, winsorization, network, and optimizer.

Ensure: Trained multi-task quantile network f_θ for one- and two-month horizons.

- 1: Transform each property’s event sequence into an event-level panel (Section III-A).
- 2: Aggregate events to ZIP×month and compute $P_{z,t}^{\text{raw}}$ via Eq. (1).
- 3: Compute county and state medians and effective index $P_{z,t}$ via Eq. (2).
- 4: Derive temporal, macro, and geo-tiling features for each ZIP×month cell.
- 5: Construct reliability weights $w_{z,t}$ using Eq. (7).
- 6: Build future labels $R_{z,t}^{(1)}, R_{z,t}^{(2)}$ as log returns; apply state-aware winsorization to obtain $\tilde{R}_{z,t}^{(1)}, \tilde{R}_{z,t}^{(2)}$.
- 7: Split labeled rows into train / validation / holdout based on calendar time.
- 8: Fit scalars and categorical vocabularies on the training set.
- 9: **for** epoch = 1 to E_{max} **do**
- 10: **for** each minibatch B from training data **do**
- 11: Encode numeric and categorical features; compute $f_\theta(B)$.
- 12: Compute quantile and median losses for both horizons.
- 13: Apply reliability weights to horizon-1 losses; sum to obtain L_{total} .
- 14: Backpropagate and update θ with AdamW.
- 15: **end for**
- 16: Evaluate validation MAE for both horizons; update best checkpoint.
- 17: **if** no improvement for patience epochs **then**
- 18: **break**
- 19: **end if**
- 20: **end for**
- 21: **return** best-performing network f_{θ^*} .

We additionally perform a **rolling 60-day backtest** with three folds, where each fold uses a distinct 60-day window as holdout and all prior data as training.

B. Error Metrics

We evaluate in **price space** (USD) by transforming predicted log returns back to index levels. For a given horizon, let P_i denote the true price level and $\hat{P}_i$ the predicted median. We compute:

- **Mean Absolute Error (MAE)**

$$\text{MAE} = \frac{1}{n} \sum_i |P_i - \hat{P}_i|;$$

- **Coefficient of Determination (R^2)**

$$R^2 = 1 - \frac{\sum_i (P_i - \hat{P}_i)^2}{\sum_i (P_i - \bar{P})^2};$$

TABLE I
MAIN HOLDOUT PERFORMANCE ON SCRAPED DATASET.

Metric	Horizon 1 (1-month)	Horizon 2 (2-month)
MAE (USD)	31.73	31.27
R^2	0.9971	0.9960
WAPE	0.0424	0.0428
MdAPE	0.0311	0.0294
Pct $ err \leq 10\%$	0.8093	0.8232
PI coverage (10–90)	0.9803	0.9774
Mean relative PI width	0.8913	0.8500

TABLE II
ROLLING 60-DAY BACKTEST ON SCRAPED DATASET.

Fold	Hor.	MAE	R^2	WAPE	MdAPE
1	1	84.83	0.9738	0.0993	0.0466
1	2	71.82	0.9837	0.0872	0.0459
2	1	39.99	0.9977	0.0474	0.0411
2	2	28.76	0.9976	0.0398	0.0341
3	1	41.54	0.9932	0.0555	0.0279
3	2	28.03	0.9978	0.0383	0.0335

- **Weighted Absolute Percentage Error (WAPE)**

$$\text{WAPE} = \frac{\sum_i |P_i - \hat{P}_i|}{\sum_i |P_i|};$$

- **Median Absolute Percentage Error (MdAPE)**

$$\text{MdAPE} = \text{median}_i \left(\frac{|P_i - \hat{P}_i|}{|P_i|} \right);$$

- **Fraction of predictions within 10%:**

$$\text{Pct}_{10} = \frac{1}{n} \sum_i \mathbf{1} \left\{ \frac{|P_i - \hat{P}_i|}{|P_i|} \leq 0.1 \right\};$$

- **Prediction interval coverage and width**, using the 10th and 90th quantiles in price space.

These metrics jointly characterize point accuracy, relative error, calibration, and sharpness of the forecast distribution.

VII. RESULTS ON SCRAPED DATASET

A. Main Holdout Performance

Table I summarizes the main holdout results for the scraped dataset, where price levels are expressed in dollars.

The two-month horizon performs slightly better or comparable to the one-month horizon across most metrics, suggesting that medium-term ZIP dynamics are smoother and easier to forecast than very short-term fluctuations.

B. Rolling 60-Day Backtest

Table II reports performance across three rolling 60-day folds, demonstrating that accuracy remains stable across time and that the model does not simply overfit to the final months.

Across all folds, the two-month horizon again matches or exceeds the one-month horizon, reinforcing the view that medium-term trajectories are more predictable than month-to-month noise.

TABLE III
OVERALL PERFORMANCE ON COMPANY INDEX DATASET (486 ROWS).

Metric	Horizon 1	Horizon 2
MAE (USD)	112.31	71.12
R^2	0.9004	0.9651
WAPE	0.6344	0.4025
MdAPE	0.9708	0.6314
Pct $ err \leq 10\%$	0.0000	0.0137
PI coverage (10–90)	0.9973	1.0000
Mean relative PI width	4.18	9.87

TABLE IV
STATE-LEVEL MAE ON COMPANY INDEX DATASET.

State	N	MAE H1 (USD)	MAE H2 (USD)
South Carolina	80	99.68	61.57
Virginia	74	153.58	92.44
Florida	67	89.90	61.90
North Carolina	55	107.44	66.90
Georgia	48	97.28	65.84
Maryland	31	128.81	83.16
West Virginia	10	105.72	62.77

VIII. RESULTS ON THIRD-PARTY COMPANY INDEX

To assess **out-of-sample generalization**, we evaluate the trained model on a separate **company index** dataset comprising 486 ZIP×month cells across seven US states. The pipeline reuses the same index construction and feature engineering logic; the only difference is that the price index is derived from a proprietary aggregation with median around \$435k.

A. Aggregate Metrics

Table III reports overall performance on this dataset.

On this more challenging out-of-sample dataset, R^2 **remains high**, especially for the two-month horizon, indicating that the model still explains the vast majority of cross-sectional variance. WAPE is higher due to the more volatile index and the fact that forecasts are transferred without retraining. Prediction intervals are **extremely conservative**, with coverage close to 100%, which is reflected in large relative widths.

B. State-Level Error Decomposition

Table IV summarizes MAE by state. **Most states exhibit two-month MAE between \$60 and \$90**, confirming that the model captures medium-term movements even when transferred geographically.

Virginia and Maryland show higher MAE, driven by a small number of high-priced, volatile ZIPs. The two-month horizon remains consistently more accurate than the one-month horizon.

C. County-Level Error Decomposition

Table V presents the top counties by number of ZIP×month cells and their mean absolute errors.

Many counties, such as **Lake, Aiken, Polk, and Spartanburg**, achieve **two-month MAE near \$60**, which is competitive for a transfer-learning setting without retraining.

TABLE V
TOP COUNTIES BY N ON COMPANY INDEX DATASET.

County	N	MAE H1	MAE H2
Wake County	29	102.88	71.65
Prince William County	26	150.59	105.85
Lake County	25	90.03	60.26
Aiken County	23	88.85	62.35
Prince Georges County	21	131.76	83.01
Brevard County	19	91.40	65.64
Spartanburg County	18	107.68	54.99
Polk County	15	82.93	59.61
Fairfax County	14	163.76	87.57
Columbia County	12	96.92	62.52

TABLE VI
REPRESENTATIVE ZIP-LEVEL MAE ON COMPANY INDEX DATASET.

ZIP code	N	MAE H1	MAE H2
29801	16	85.15	67.35
30102	14	95.69	64.66
27587	13	89.07	72.48
32796	10	93.81	62.94
33851	10	81.34	61.75
22026	10	133.60	102.78
29909	9	124.56	78.66
30814	8	95.27	63.43
25438	8	106.31	62.24
20112	7	157.29	111.37

D. ZIP-Level Error Decomposition

Finally, Table VI displays representative ZIP codes with the largest sample sizes.

Most ZIPs with sufficient support exhibit two-month MAE between \$60 and \$80, confirming that the model’s learned representation of momentum, seasonality, and macro effects generalizes beyond the original scraped sample.

IX. DISCUSSION AND FUTURE WORK

The proposed ZIP×month TAB-MTL framework demonstrates that a carefully engineered tabular network can deliver highly competitive spatio-temporal forecasts without relying on complex sequence models or graph neural networks. Key empirical observations include:

- **Two-month horizons are more predictable than one-month horizons**, both in-sample and out-of-sample, likely reflecting the smoother dynamics of medium-term price movements.
- **Reliability weighting and winsorization are crucial** for stabilizing training in the presence of sparse ZIPs and extreme month-to-month jumps.
- **The model transfers reasonably well to a third-party company index**, preserving high R^2 and manageable MAE despite differences in geography and data collection.

Future work will explore:

- integrating true H3 geo-indices and explicit neighbor graph structure;
- combining the ZIP×month index model with property-level hedonic models in a fully hierarchical stack;

- augmenting macro features with local supply indicators such as permits and inventory.

Ultimately, our goal is a unified, multi-scale framework that simultaneously forecasts market trajectories and individual property deviations, enabling more informed, data-driven decisions for builders, lenders, and investors.

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